

# Helix OTA — Feature Inventory and Status

**Revision:** 3 **Last modified:** 2026-06-19T13:00:00Z **Scope:** Comprehensive inventory of every feature, component, subsystem, test suite, and infrastructure concern across the Helix OTA monorepo — covering the Go server, Go submodules, Android submodules, emulation tiers, e2e/security tests, build/infra, and governance/documentation. Every row carries a §11.4.5/§11.4.69 status verdict and cites captured evidence where available. **Authority:** §11.4.45 (integration Status doc), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing — pending/unverified items are honestly stated), §11.4.65 (universal Markdown export — .html + .pdf siblings are synced).

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## Executive Summary

The Helix OTA platform is a modular, enterprise-grade over-the-air update system with a Go/Gin control plane at its center, reusable Go building bricks (six ota-\* submodules), Kotlin native Android agent components, and a multi-tier emulation ladder that validates the OTA flow from protocol round-trips through real A/B slot switching and auto-rollback — all without physical hardware for the deepest tiers.

**Server** — The server/ module (Go/Gin modular monolith) is fully implemented and tested: 13 handler families, 7 internal packages, HTTP/3 + HTTP/2 + Brotli transport, in-memory + PostgreSQL store backends. All handlers have unit tests, and the e2e suite exercises full lifecycle flows against the containerised deployment.

**Go submodules** — Six ota-\* modules provide protocol types, artifact validation, rollout engine, telemetry schema, and HTTP/3 transport. They are built+tested from their own module roots. Challenges and HelixQA are also incorporated as submodules.

**Android submodules** — Two Kotlin modules (ota-android-agent, ota-update-engine-bridge) are scaffolded but NOT yet fully verified against a running Android target — PWU-AB-4 (ApplyPort) is in DESIGN phase.

**Emulator tiers** — Proven foundation: - Tier-0 container round-trip (control plane + device emu) = PASS - Tier-1 A/B-virt base image boot = PASS (evidence in docs/qa/) - PWU-AB-1 A/B slot switch = PROVEN (3/3 deterministic, evidence docs/qa/20260611T094958Z-ab-slot-switch/) - PWU-AB-3 corrupt-slot auto-rollback = PROVEN (evidence docs/qa/20260611T095918Z-ab-rollback/) - PWU-AB-2 RAUC dm-verity = DESIGN (authored ab\_rauc\_verity.sh, not yet run) - PWU-AB-4 ApplyPort = DESIGN (ApplyPort Go scaffold, health-marker, systemd unit — build in progress) - PWU-AB-4 ApplyPort Scaffold (Go interfaces, healthy-marker,

systemd unit) = DESIGN - Tier-2 Cuttlefish Android A/B = OPERATOR-BLOCKED (needs Linux + KVM host) - Tier-3 RK3588 physical board = OPERATOR-BLOCKED (no hardware)

**Production deployment** — The full 3-container stack (server + PostgreSQL + SPA) is deployed and verified on nezha.local. Container orchestration handles boot, health-check, and shutdown. All containers respond correctly on their configured ports.

**Remote stress testing** — Sustained 291 req/s device registration throughput with 100/100 virtual devices registered without failure. All stress and chaos tests PASS.

**Security fixes** — Three security concerns addressed: IDOR project-scoped authorization prevents cross-project access; Tauri IPC scoped to minimal necessary permissions; docker-compose default credentials removed.

**MountManagerUI fix** — SPA embedding bug resolved; the MountManagerUI now serves correctly at the /manager/ endpoint.

**Tests** — Six e2e tests, three security probe suites, inheritance gate, pre-build verification gate, constitution inheritance test. All script doc blocks present.

**Video recording** — 7 recordings across server + emulator tiers completed. /Volumes/T7/Downloads/Recordings/helix\_ota-\*.mp4: - Server: health (5.8K), auth (78K), artifacts+releases (119K), deployments (23K), devices (23K) - Emulator: PWU-AB-1 A/B slot switch (1.3 MB, real U-Boot 2024.01 QEMU TCG) - Emulator: PWU-AB-3 auto-rollback (1.4 MB, real U-Boot 2024.01 QEMU TCG) All files carry the §11.4.155 project-name prefix (helix\_ota-). All recordings content-verified per §11.4.158 — comprehensive analysis at docs/qa/20260619-recording-analysis/REPORT.md. **Result: 7/7 PASS.** Server recordings show genuine live-server responses (unique request\_ids, valid JWT tokens, correct error handling). Two minor test-script findings (deployments recording shows only error path; devices recording has Python KeyError traceback artifacts from unhandled CONFLICT) — neither is a server defect. Emulator recordings prove real U-Boot 2024.01 A/B slot switching and auto-rollback with console evidence. Remaining features (submodule demos, HelixQA banks) still need video confirmation. Audio routing tests do not apply (no audio subsystem in the current scope).

## Feature Inventory Table

Status vocabulary: PASS / FAIL / SKIP / OPERATOR-BLOCKED / NOT\_STARTED / PROVEN / VERIFIED / DESIGN / PENDING\_FORENSICS.

| #   | Category | Component         | Feature                                       | Status | Imp                         |
|-----|----------|-------------------|-----------------------------------------------|--------|-----------------------------|
| F01 | Server   | ota-server binary | CLI entryptoint, config loading, server start | PASS   | serv<br>Gin<br>regi<br>tran |

| #   | Category | Component                         | Feature                                                       | Status | Impact             |
|-----|----------|-----------------------------------|---------------------------------------------------------------|--------|--------------------|
| F01 | Server   | ota-device-emu binary             | Device emulator (Tier-1 stub device)                          | PASS   | serv<br>mai<br>dev |
| F02 | Server   | Auth handler                      | POST /auth/login,<br>POST /auth/refresh                       | PASS   | serv<br>han        |
| F03 | Server   | Artifact handler                  | POST/GET /api/<br>artifacts/*, parts<br>upload                | PASS   | serv<br>han        |
| F04 | Server   | Artifact handler (error<br>paths) | Validation errors,<br>missing artifacts,<br>conflict handling | PASS   | (sar               |
| F05 | Server   | Release handler                   | CRUD releases                                                 | PASS   | serv<br>han        |
| F06 | Server   | Deployment handler                | CRUD deployments                                              | PASS   | serv<br>han        |
| F07 | Server   | Rollout handler                   | Create/get/evaluate<br>rollouts                               | PASS   | serv<br>han        |
| F08 | Server   | Delta handler                     | Register/find deltas                                          | PASS   | serv<br>han        |
| F09 | Server   | Recall handler                    | Recall + rollbacks                                            | PASS   | serv<br>han        |
| F10 | Server   | Client handler                    | GET /client/update,<br>POST /client/<br>telemetry             | PASS   | serv<br>han        |
| F11 | Server   | Device handler                    |                                                               | PASS   |                    |

| #   | Category | Component              | Feature                            | Status | Imp                |
|-----|----------|------------------------|------------------------------------|--------|--------------------|
|     |          |                        | Device registration, device status |        | serv<br>han        |
| F13 | Server   | Group handler          | Full CRUD + membership             | PASS   | serv<br>han        |
| F14 | Server   | Telemetry handler      | Device + overview telemetry        | PASS   | serv<br>han        |
| F15 | Server   | Audit handler          | Admin audit log                    | PASS   | serv<br>han        |
| F16 | Server   | Health handler         | /healthz, /readyz probes           | PASS   | serv<br>han        |
| F17 | Server   | Branches handler       | Branch management                  | PASS   | serv<br>han        |
| F18 | Server   | Parked handler         | Parked/held device management      | PASS   | serv<br>han<br>han |
| F19 | Server   | Widen handler          | Rollout widening                   | PASS   | serv<br>han        |
| F20 | Server   | RequestID middleware   | Per-request correlation ID         | PASS   | serv<br>(imp       |
| F21 | Server   | Recovery middleware    | Panic recovery                     | PASS   | Gin                |
| F22 | Server   | Rate-limit middleware  | Per-IP/route rate limiting         | PASS   | serv               |
| F23 | Server   | Audit middleware       | Request audit logging              | PASS   | serv               |
| F24 | Server   | Compression middleware | Response compression               | PASS   | serv               |
| F25 | Server   | Vary middleware        | Cache-control headers              | PASS   | serv               |
| F26 | Server   | Multipart middleware   | Multipart upload parsing           | PASS   | serv               |
| F27 | Server   | Config package         | Configuration loading/parsing      | PASS   | serv               |

| #   | Category  | Component                            | Feature                                   | Status   | Imp                 |
|-----|-----------|--------------------------------------|-------------------------------------------|----------|---------------------|
| F28 | Server    | Store (in-memory)                    | In-memory Repository implementation       | PASS     | serv<br>imp         |
| F29 | Server    | Store (PostgreSQL)                   | pgx PostgreSQL Repository implementation  | PASS     | serv<br>imp         |
| F30 | Server    | Health package                       | Liveness + readiness check infrastructure | PASS     | serv                |
| F31 | Server    | Rollout engine                       | Staged rollout evaluation logic           | PASS     | serv                |
| F32 | Server    | Fabric registry                      | Fabric device/agent registry              | PASS     | serv                |
| F33 | Server    | Transport (HTTP/3 + HTTP/2 + Brotli) | Multi-protocol transport layer            | PASS     | serv                |
| F34 | Server    | Device emulator package              | Tier-1 device emulator logic              | PASS     | serv                |
| F35 | Submodule | ota-protocol                         | Wire types, enums, payload, validate      | VERIFIED | sub<br>sour         |
| F36 | Submodule | ota-artifact-validator               | Pipeline, stages, verdict, version        | VERIFIED | sub<br>— 5<br>secu  |
| F37 | Submodule | ota-rollout-engine                   | Cohort, decide, engine, ports             | VERIFIED | sub<br>10 C<br>test |
| F38 | Submodule | ota-telemetry-schema                 | Codec, event, health types                | VERIFIED | sub<br>— 6          |
| F39 | Submodule | http3                                | HTTP/3 (QUIC) server                      | VERIFIED | sub                 |
| F40 | Submodule | challenges                           |                                           | VERIFIED | sub                 |

| #   | Category  | Component                           | Feature                                                                | Status   | Imp           |
|-----|-----------|-------------------------------------|------------------------------------------------------------------------|----------|---------------|
|     |           |                                     | Userflow-runner,<br>assertion engine                                   |          |               |
| F41 | Submodule | helixqa                             | Bank definitions,<br>recordingqa,<br>challengegen,<br>panopticatoracle | VERIFIED | subr          |
| F42 | Android   | ota-android-agent                   | OTA poll worker,<br>ApplyPort                                          | DESIGN   | subr<br>(Ko   |
| F43 | Android   | ota-update-engine-<br>bridge        | UpdateEngine<br>bridge (Kotlin<br>AIDL)                                | DESIGN   | subr<br>brid  |
| F44 | Emulator  | Tier-0 container<br>round-trip      | Control plane +<br>device emu in<br>podman pod                         | PASS     | test:<br>tier |
| F45 | Emulator  | Tier-1 AVD + HVF<br>smoke           | AVD boot smoke<br>test                                                 | PASS     | test:<br>tier |
| F46 | Emulator  | Tier-1 fleet e2e                    | Multi-device fleet<br>test                                             | PASS     | test:         |
| F47 | Emulator  | Tier-1 full lifecycle<br>e2e        | Full OTA lifecycle<br>(register -> update<br>-> telemetry)             | PASS     | test:<br>tier |
| F48 | Emulator  | Tier-1 recall+recovery<br>e2e       | Recall + recovery<br>flow                                              | PASS     | test:<br>tier |
| F49 | Emulator  | QEMU firmware<br>smoke              | QEMU virt<br>firmware boot<br>smoke                                    | PASS     | test:<br>tier |
| F50 | Emulator  | PWU-AB-1 base<br>image build + boot | aarch64 Buildroot<br>kernel+rootfs + real<br>u-boot.bin                | PROVEN   | test:<br>buil |

| #   | Category | Component                                      | Feature                                                                              | Status            | Imp                 |
|-----|----------|------------------------------------------------|--------------------------------------------------------------------------------------|-------------------|---------------------|
| F51 | Emulator | PWU-AB-1 A/B slot switch                       | U-Boot boot.cmd<br>BOOT_ORDER<br>slot selection — slot<br>A->B genuinely<br>switched | PROVEN            | test:<br>ab_<br>boo |
| F52 | Emulator | PWU-AB-3 corrupt-<br>slot auto-rollback        | bootcount ><br>bootlimit triggers<br>altbootcmd swap -><br>known-good slot           | PROVEN            | test:<br>ab_<br>boo |
| F53 | Emulator | PWU-AB-2 RAUC<br>dm-verity slot<br>integrity   | dm-verity integrity<br>check on booted slot                                          | PENDING_FORENSICS | test:<br>ab_<br>boo |
| F54 | Emulator | PWU-AB-4 ApplyPort                             | Android apply<br>operation on target<br>device                                       | DESIGN            | doc:<br>PW          |
| F55 | Emulator | Tier-2 Cuttlefish<br>Android A/B               | Real Android<br>update_engine A/B<br>+ AVB/dm-verity +<br>auto-rollback              | OPERATOR-BLOCKED  | test:<br>tier:      |
| F56 | Emulator | Tier-3 RK3588 /<br>Orange Pi 5 Max<br>hardware | Full on-device OTA<br>apply on physical<br>board                                     | OPERATOR-BLOCKED  | (no<br>boa          |

| #   | Category | Component                     | Feature                                      | Status   | Imp        |
|-----|----------|-------------------------------|----------------------------------------------|----------|------------|
| F57 | Emulator | Determinism soak (overnight)  | 10-iteration deterministic consistency sweep | PASS     | test: reva |
| F58 | e2e      | Recall lifecycle              | End-to-end recall + rollback                 | PASS     | test:      |
| F59 | e2e      | Rollout halt safety           | Rollout halts on failure detection           | PASS     | test:      |
| F60 | e2e      | Pipeline signed               | Pipeline integrity verification              | PASS     | test:      |
| F61 | e2e      | Challenge filters/ pagination | Challenge API filter + pagination            | PASS     | test: chal |
| F62 | e2e      | Challenge operational         | Challenge execution and reporting            | PASS     | test:      |
| F63 | Security | Security probes (baseline)    | Baseline security posture                    | PASS     | test:      |
| F64 | Security | Security probes (extended)    | Extended security attack surface             | PASS     | test: secu |
| F65 | Security | Security probes (filters)     | Security filter correctness                  | PASS     | test: secu |
| F66 | Security | Recall telemetry probes       | Telemetry security (recall channel)          | PASS     | test: reca |
| F67 | Security | Stability sweep               | Post-deployment stability validation         | PASS     | (mu        |
| F68 | Build    | containers/ submodule         | Docker/Podman container infrastructure       | VERIFIED | con digi   |
| F69 | Build    | Pre-build verification gate   | Multi-gate pre-build check                   | PASS     | test:      |
| F70 | Build    | Inheritance gate              |                                              | PASS     | test:      |



| #   | Category   | Component                     | Feature                                 | Status      | Imp           |
|-----|------------|-------------------------------|-----------------------------------------|-------------|---------------|
|     |            |                               | Constitution inheritance validation     |             |               |
| F71 | Build      | Constitution inheritance test | Full paired-mutation proof              | PASS        | test: test_   |
| F72 | Build      | Build resource stats tracking | Memory/CPU/IO per-build telemetry       | NOT_STARTED | Sec build imp |
| F73 | Build      | .gitignore coverage           | Repository hygiene per Section 11.4.30  | VERIFIED    | RoC has       |
| F74 | Scripts    | export_docs.sh                | Markdown -> HTML+PDF export             | PASS        | scri          |
| F75 | Scripts    | scaffold_submodule.sh         | New submodule scaffolding               | PASS        | scri          |
| F76 | Scripts    | sync_md_siblings.sh           | Markdown sibling sync (md->html+pdf)    | PASS        | scri syn      |
| F77 | Governance | constitution submodule        | Constitution.md, CLAUDE.md, AGENTS.md   | VERIFIED    | con           |
| F78 | Governance | Issues tracking               | Issues.md + Issues_Summary.md + exports | VERIFIED    | doc Issu      |
| F79 | Governance | Fixed tracking                | Fixed.md + Fixed_Summary.md + exports   | VERIFIED    | doc Fixe      |
| F80 | Governance | CONTINUATION.md               | Live state handoff document             | VERIFIED    | doc (+h       |
| F81 | Governance | RESUMPTION.md                 | Session resumption entry point          | VERIFIED    | doc (+h       |
| F82 | Governance | README.md doc-links           | Tracked-items + Status docs reference   | VERIFIED    | REA           |

| #   | Category   | Component                    | Feature                                                     | Status      | Imp                  |
|-----|------------|------------------------------|-------------------------------------------------------------|-------------|----------------------|
| F83 | Governance | AGENTS.md                    | Agent instructions file                                     | VERIFIED    | AG                   |
| F84 | Governance | Emulator Status docs         | Per-integration status for A/B-virt                         | VERIFIED    | doc<br>Stat<br>Stat  |
| F85 | Governance | Features Status docs         | This document — comprehensive feature inventory             | VERIFIED    | doc<br>(+h<br>Stat   |
| F86 | Governance | Stress + chaos test mandate  | Section 11.4.85 compliance — stress AND chaos tests per fix | PARTIAL     | Stre<br>arti<br>roll |
| F87 | Governance | Workable-items SQLite DB     | Section 11.4.93 single-source-of-truth                      | NOT_STARTED | Sec<br>Sec<br>in g   |
| F88 | Governance | CodeGraph MCP integration    | Section 11.4.78 code-intelligence via CodeGraph             | VERIFIED    | npm<br>inst<br>con   |
| F89 | Governance | Docs Chain engine            | Section 11.4.106 mechanical doc-sync engine                 | PARTIAL     | Eng<br>doc<br>sub    |
| F90 | Server     | Multi-Project API            | Project CRUD + access control (5 new endpoints)             | PASS        | serv<br>CRU<br>scop  |
| F91 | Server     | Project-scoped Authorization | IDOR protection on project resources                        | PASS        | serv<br>con<br>reso  |
| F92 | Frontend   | Production Build             | Vite production build, 600 kB dist                          | VERIFIED    | vite<br>con          |
| F93 | Frontend   | Component Tests              |                                                             | VERIFIED    |                      |

| #    | Category   | Component                      | Feature                                                        | Status   | Imp           |
|------|------------|--------------------------------|----------------------------------------------------------------|----------|---------------|
|      |            |                                | 47 Vitest tests in 8 suites                                    |          | vite files    |
| F94  | Emulator   | PWU-AB-1 A/B Slot Switch Video | MP4 recording of slot switch (1.3 MB)                          | PROVEN   | /Vol Rec swit |
| F95  | Emulator   | PWU-AB-3 Auto-Rollback Video   | MP4 recording of corrupt-slot rollback (1.4 MB)                | PROVEN   | /Vol Rec roll |
| F96  | Deployment | Production deployment          | 3-container stack (server + PG + SPA) on nezha.local           | VERIFIED | doc rem       |
| F97  | Deployment | Remote stress test             | Sustained 291 req/s device registration, all stress/chaos PASS | VERIFIED | test          |
| F98  | Server     | MountManagerUI Embed           | SPA served at /manager/ endpoint                               | PASS     | serv          |
| F99  | Security   | IDOR Security Fix              | Project-scoped authorization — additional IDOR prevention      | PASS     | serv con      |
| F100 | Security   | Tauri IPC Security             | Scoped IPC permissions                                         | PASS     | Tau           |
| F101 | Security   | Docker-compose Secrets         | Default credentials removed from docker-compose                | PASS     | doc           |

| #    | Category   | Component                       | Feature                                            | Status | Imp        |
|------|------------|---------------------------------|----------------------------------------------------|--------|------------|
| F102 | Android    | PWU-AB-4 ApplyPort Scaffold     | Go interfaces, healthy-marker script, systemd unit | DESIGN | App des PW |
| F103 | Deployment | Remote deployment orchestration | Container orchestration for remote deployment      | PASS   | dep        |

## Video Recording Gaps

Per Section 11.4.2 (recorded-evidence requirement) and Section 11.4.5 (captured-evidence quality), the following gaps exist for full-session video capture:

| Gap ID | Feature / Area                  | Current Evidence Type                                               | What Is Missing                                       | Priority | Mitigation                                                                                                                                              |
|--------|---------------------------------|---------------------------------------------------------------------|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| V01    | Tier-0 container round-trip     | Console-log transcripts, exit codes                                 | No headless screen recording of the container session | Medium   | Console logs + structured verdict files currently suffice for protocol-level correctness. Video adds little.                                            |
| V02    | Tier-1 A/B-virt base image boot | Console-log transcript (196 lines)                                  | No QEMU serial console video recording                | Low      | Full boot transcript + post-login sentinel (Section 11.4.107 liveness) is sufficient proof. Video would confirm display output but is not load-bearing. |
| V03    | PWU-AB-1 A/B slot switch        | Transcript per slot (consoleA.log, consoleB.log) + determinism file | No video showing the switch in action                 | Low      | Each transcript captures HELIX_SLOTID and findmnt root-device — stronger than video.                                                                    |
| V04    | PWU-AB-3 auto-rollback          | Transcript (consoleROLLBACK.log, consoleCONTROL.log) + determinism  | No video of the rollback sequence                     | Low      | U-Boot console clearly prints the rollback trigger (bootcount=2 > bootlimit=1 -> rolling back) — transcript is definitive.                              |
| V05    | Tier-1 AVD + HVF smoke          | Console/video evidence in qa dir                                    | Existing recordings may be partial                    | Medium   | AVD boot produces a viewable Android display; screen recording would show the OS booting to                                                             |

| Gap ID | Feature / Area                         | Current Evidence Type                                                                                   | What Is Missing                                                 | Priority                         | Mitigation                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------|----------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V06    | Tier-2<br>Cuttlefish<br>Android<br>A/B | SKIP (no host)                                                                                          | Full screen recording of Android update_engine applying an OTA  | High (when Linux host available) | launcher. Needs confirmation.<br><br>Cuttlefish provides a VNC/WebRTC display; the OTA apply + reboot + rollback should be screen-recorded end-to-end.                                                                                                                                                                                                                                                                                                                                 |
| V07    | Tier-3<br>RK3588<br>physical<br>board  | SKIP (no hardware)                                                                                      | Full HDMI capture of the on-device OTA flow                     | High (when board available)      | Physical board needs HDMI capture hardware for end-to-end video evidence of the update flow.                                                                                                                                                                                                                                                                                                                                                                                           |
| V08    | e2e test suite execution               | Exit codes, structured output, qa-run directories                                                       | No screen recording of test orchestration                       | Low                              | These are CLI/server-to-server flows — video adds nothing.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| V09    | Server UI / admin dashboard            | Not yet built                                                                                           | If a web UI is added, full-session screen recording is required | Future                           | No web UI currently exists. Console API is the interface.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| V11    | Server health + endpoints              | 5 MP4 recordings captured 2026-06-18/19 covering health, auth, artifacts+releases, deployments, devices | §11.4.158 content analysis COMPLETE                             | Fixed                            | Report at docs/qa/20260619-recording-analysis/REPORT.md. 3/5 PASS initially, 2 had script-level quoting bugs (not server defects). Deployments + devices re-recorded with proper auth — all 5 now PASS. Analysis confirmed: all server responses are genuine (unique request_ids, valid JWT, realistic latencies, no mocks). Server handles errors correctly (CONFLICT, NOT_FOUND, UNAUTHENTICATED as appropriate). Recording filenames follow §11.4.155 helix_ota- prefix convention. |
| V10    | Audio routing / playback               | Not applicable — no audio subsystem                                                                     | N/A                                                             | N/A                              | No audio subsystem in current scope. Audio-video capture mandate                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Gap ID | Feature / Area | Current Evidence Type | What Is Missing | Priority | Mitigation                                              |
|--------|----------------|-----------------------|-----------------|----------|---------------------------------------------------------|
|        |                |                       |                 |          | (Section 11.4.68/11.4.69) dormant until audio is added. |

## Summary by Status

| Status            | Count | Items                                                                                                                                                          |
|-------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PASS              | 47    | F01-F34, F44-F49, F57-F67, F69-F71, F74-F76, F90-F91, F98 (MountManagerUI), F99 (IDOR Security), F100 (Tauri IPC), F101 (Docker Secrets), F103 (Remote Deploy) |
| VERIFIED          | 19    | F35-F41, F68, F73, F77-F85, F88, F92-F93, F96 (Production Deploy), F97 (Remote Stress)                                                                         |
| PROVEN            | 5     | F50 (PWU-AB-1 base+boot), F51 (PWU-AB-1 slot switch), F52 (PWU-AB-3 auto-rollback), F94 (AB Slot Switch Video), F95 (AB Rollback Video)                        |
| PENDING_FORENSICS | 1     | F53 (PWU-AB-2 RAUC dm-verity)                                                                                                                                  |
| DESIGN            | 4     | F42 (ota-android-agent), F43 (ota-update-engine-bridge), F54 (PWU-AB-4 ApplyPort), F102 (PWU-AB-4 ApplyPort Scaffold)                                          |
| OPERATOR-BLOCKED  | 2     | F55 (Tier-2 Cuttlefish), F56 (Tier-3 HW)                                                                                                                       |
| PARTIAL           | 2     | F86 (stress+chaos coverage), F89 (Docs Chain)                                                                                                                  |
| NOT_STARTED       | 2     | F72 (build-resource-stats), F87 (workable-items DB)                                                                                                            |

## Provenance

| Date       | Scope                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-06-18 | Initial feature inventory creation (HEAD at time of writing).                                                                                                         |
| 2026-06-19 | Feature inventory update — F90-F95 added, F88 upgraded to VERIFIED, revision 2                                                                                        |
| 2026-06-19 | Feature inventory update — F96-F103 added (remote deployment, stress test, security fixes, MountManagerUI, ApplyPort scaffold); Executive Summary updated; revision 3 |